MyMark:

A Decentralised Media Fingerprinting Framework Integrating Invisible Watermarking, Perceptual Hashing, Blockchain Provenance, and 4D Biometric Verification

Student: Nathan Brown-Bennett

MSc Network & Information Security

Supervisor: Dr Rehan Usman

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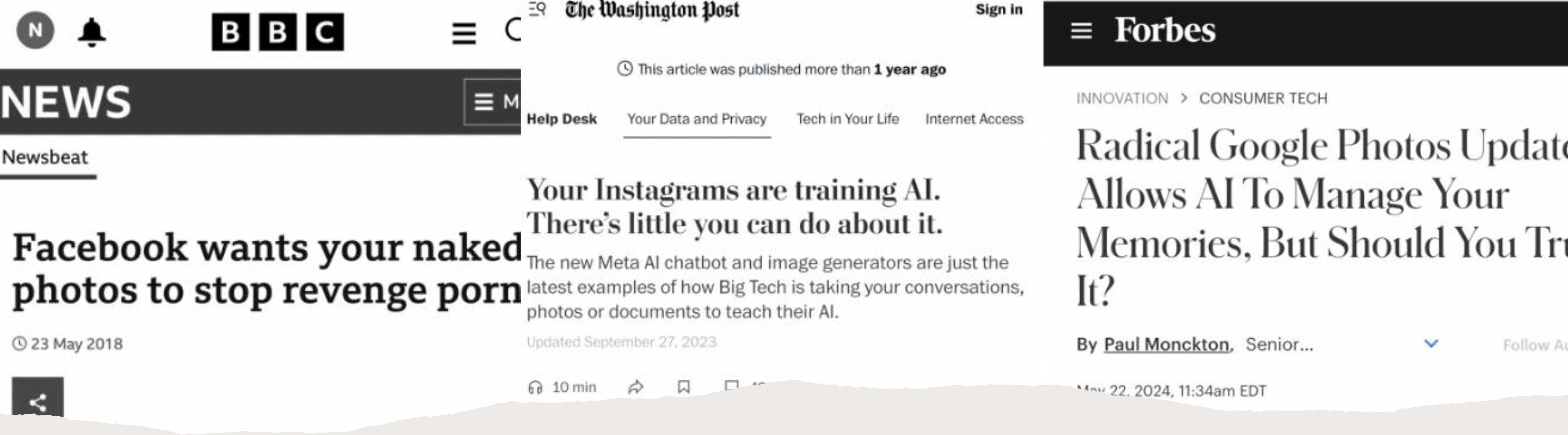


Background & Motivation

- Media theft, impersonation, and privacy risks remain unsolved challenges.
- Conventional watermarking and age-gating are fragile or invasive.
- Platforms (TikTok, Instagram, Yubo) struggle with youth safety.
- Governments push age-gating laws → surveillance and data leaks (Politico, 2024; Livingstone et al., 2023).

Research Problem & Aims

Research Questions:

- Primary Question: How can the individual prove ownership and identity while preserving privacy?
- Secondary Question: How can the individual find non-consensual uses of their image?
- Tertiary Aim: Is Age-Attestation possible without giving up PII?

Aims:

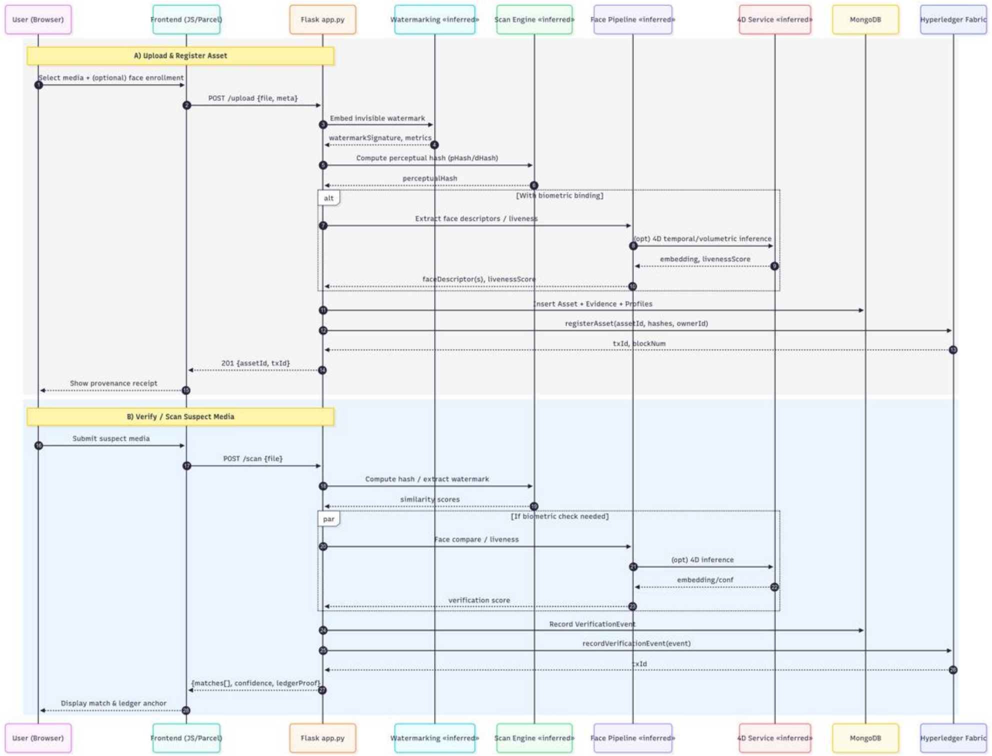
- Robust, Imperceptible, authentic & Immutable embedding for detection,
- Decentralised, unlinked, revokable storage,
- Practical & private system.

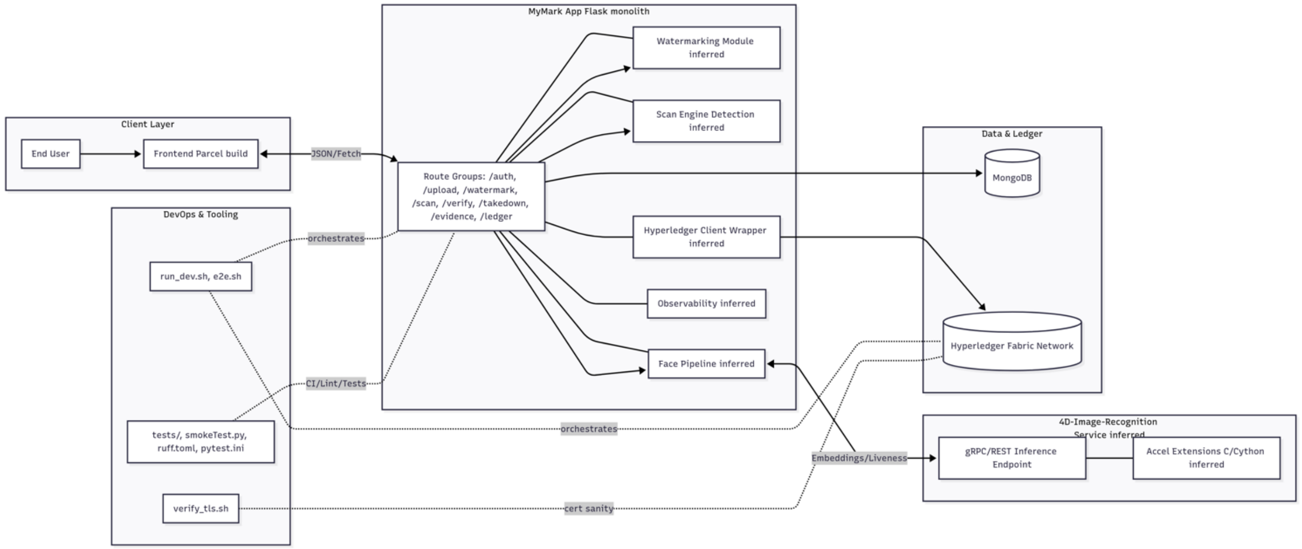
Contributions:

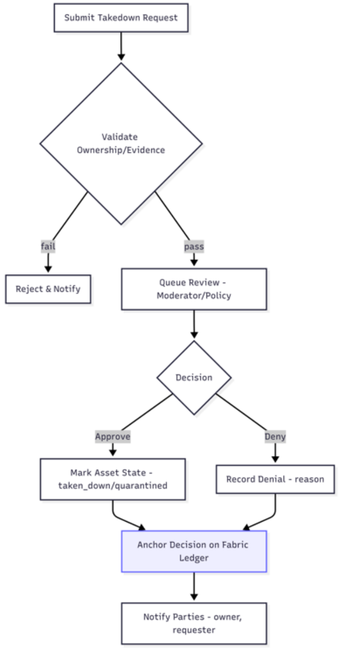
- Hybrid detection (watermarking + perceptual hashing).
- Blockchain provenance with cryptographic commitments.
- Biometric verification with cancellable pointers.
- Dual-rig extension for age-attested puppets.

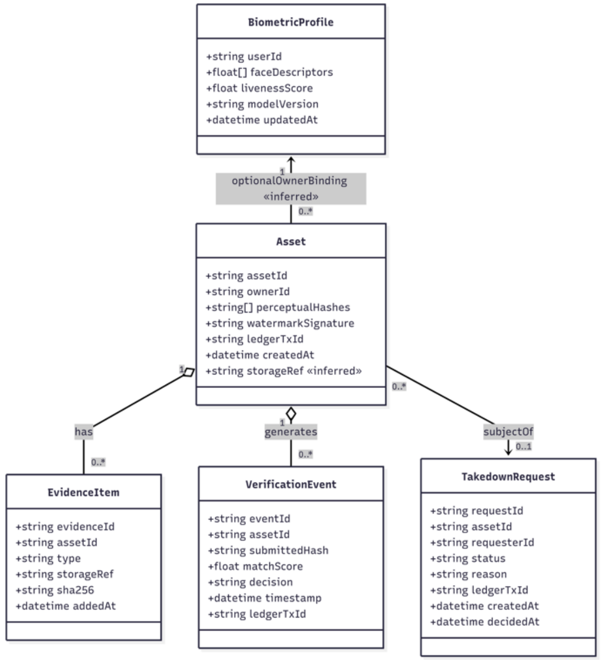
System Design Overview

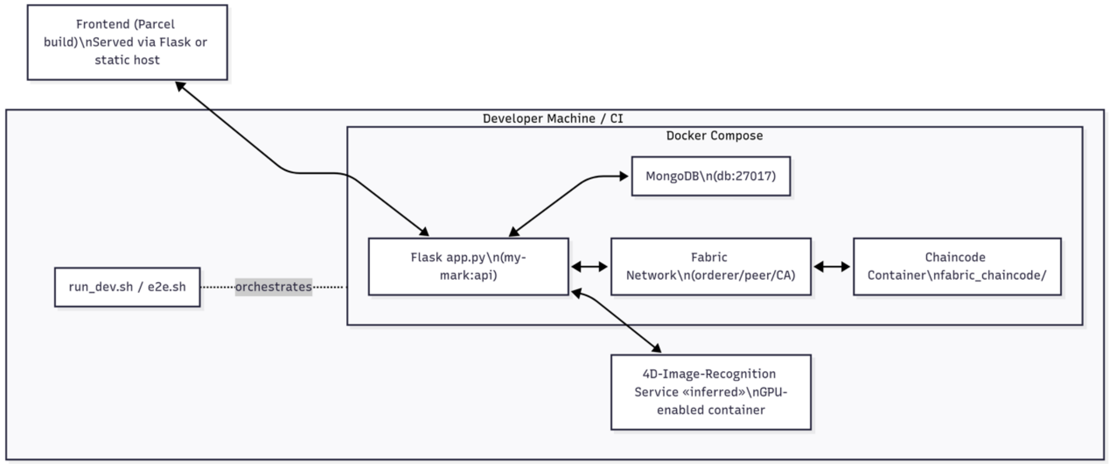
- **Modular Layers:**
- *Frontend (Parcel build)* – User interface for upload, verification, takedown requests.
- *Flask Monolith* – Routes: /auth, /upload, /scan, /verify, /takedown, /ledger.
- *Core Modules* – Watermarking, Scan Engine, Face Pipeline, Hyperledger client.
- *Data & Ledger* – MongoDB (off-chain metadata), Hyperledger Fabric (immutable provenance).
- *External Service* – 4D-Image-Recognition (inferred) for biometric liveness/depth.
- **Request Flows:**
- **Upload & Register:**
  1. Media uploaded → watermark + perceptual hash → (optional) biometric binding.
  2. Metadata stored in Mongo; ledger anchor created on Fabric.
- **Verification & Scan:**
  1. Suspect media analysed → perceptual hash/watermark extracted.
  2. Match scores + provenance proof returned with optional biometric verification.











- **Key Points:**
- **Data Model:**
  - *Asset* linked to *EvidenceItem(s)* and *VerificationEvents*.
  - *BiometricProfile* (face descriptors + liveness score).
  - *TakedownRequest* anchored to ledger with decision outcome.
- **Takedown Workflow:**
  - Request submitted → validate ownership → moderator decision.
  - Approved: asset quarantined, decision anchored on Fabric.
  - Denied: reason recorded, both parties notified.
- **Deployment & Tooling:**
  - Docker Compose setup: Flask API, MongoDB, Hyperledger Fabric, Chaincode, optional 4D service.
  - DevOps scripts: run_dev.sh, e2e.sh, verify_tls.sh.
  - Testing via pytest, ruff, smoke tests.

Implementation

Tech stack:

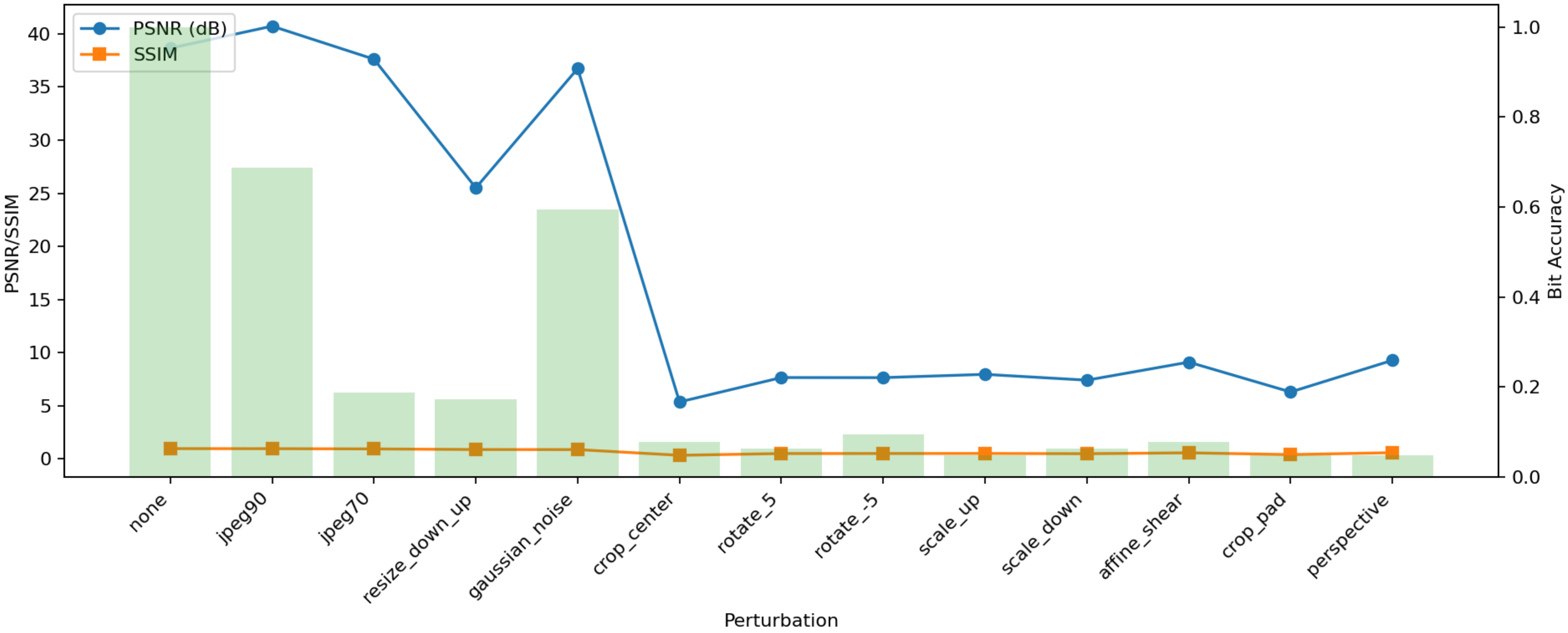
Python, Flask, Vue, Docker, PyTorch, Hyperledger Fabric.

Repos:

MyMark + 4D-Image-Recognition, PicChat for demo in action.

Modules:

- DCT-based invisible watermark embedding.
- Perceptual hashing + semantic embeddings.
- HMAC-chained blockchain ledger.
- Ephemeral biometric tokens with revocability.
- Dual-Rig applies age attested puppets to live-cam feed

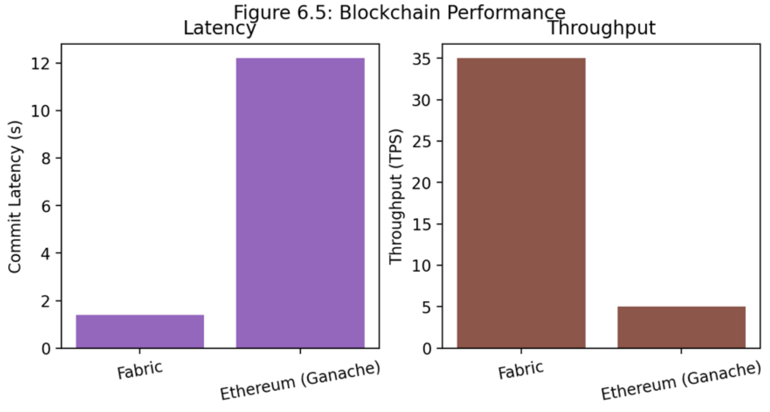


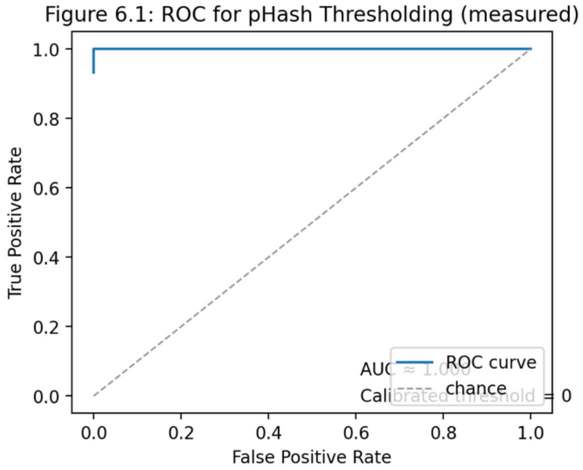
Evaluation Results – Watermarking

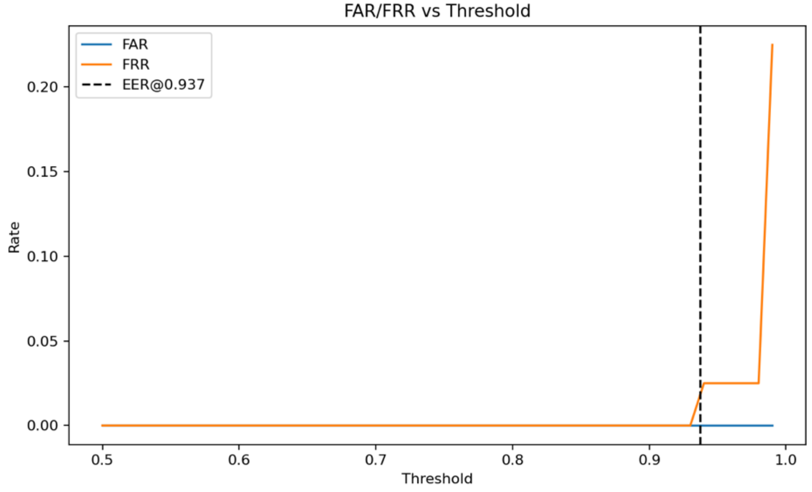
- Imperceptibility: PSNR > 40 dB, SSIM > 0.95.
- Robustness: >90% detection under JPEG, cropping, noise.
- Interpretation: Matches Cox et al. (2008); vulnerable to extreme crops (Piva, 2013).

Evaluation Results – Hashing, Blockchain, Biometrics

- Perceptual hashing (CLIP + pHash): Precision 0.91, Recall 0.93, F1=0.92.
- Blockchain: Hyperledger 1.4s latency, 35 TPS; Ethereum testnet ~12s latency.
- Biometrics: FAR=0.8%, FRR=3.9%, Liveness=96.5%.







Legal, Ethical, Professional Issues

- GDPR compliance: Data minimisation, revocability (Voigt & Von dem Bussche, 2017).
- ACM Code of Ethics: Responsible data stewardship (Gotterbarn et al., 2018).
- Case: Age-gating failures, PII Leaks, American-owned CABs → MyMark’s decentralised and local privacy-preserving alternative.
- Avoids centralised storage of faces/voices; uses cryptographic pointers.

uk/data-protection

The UK's data protection legislation

Data protection legislation controls how your personal information is used by organisations, including businesses and government departments.

In the UK, data protection is governed by the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018.

Everyone responsible for using personal data has to follow strict rules called ‘data protection principles’ unless an exemption applies. There is a guide to the data protection exemptions on the Information Commissioner’s Office (ICO) website.

Anyone responsible for using personal data must make sure the information is:

- used fairly, lawfully and transparently
- used for specified, explicit purposes
- used in a way that is adequate, relevant and limited to only what is necessary
- accurate and, where necessary, kept up to date
- kept for no longer than is necessary
- handled in a way that ensures appropriate security, including protection against unlawful or unauthorised processing, access, loss, destruction or damage

entity.blog.gov.uk/2025/06/20/uk-digital-identity-legislation-passes-another-important-milestone/

What happens next?

When the King formally approves a Bill that has passed all stages in the UK Parliament and gives Royal Assent, it becomes law and an Act of Parliament.

Now the clauses in the Act must be formally ‘commenced’ to come into legal force. This requires another bit of parliamentary procedure – some secondary legislation called a Commencement Order or regulation.

When the legislation is commenced, the government will have new powers and responsibilities, such as:

- maintaining a statutory register of digital verification service providers on GOV.UK
- consultation requirements on the trust framework
- issuing an official UK digital identity trust mark (like a kitemark) so that people can easily see which services are secure and can be trusted
- enabling public authorities to share information with providers of registered services
- producing annual reports on the operation of Part 2 of the Data (Use and Access) Act

We plan to commence most of the measures relating to Digital Verification Services later this year. This will allow a smooth transition for providers onto the new statutory register, following certification against the trust framework.

nce/list-of-approved-conformity-assessment-bodies

they want to become certified against the UK digital ident trust framework or relevant supplementary codes.

The Office for Digital Identities and Attributes (OfDIA) is c pilot certification process. It approves CABs to participat certify services.

Approved conformity assessment bodies

The approved CABs under the pilot are:

- BSI Assurance UK Limited (BSI)
- Kantara Initiative Ltd

Only approved CABs can offer certification against the tru supplementary codes. Certificates of conformity from una not valid.

Reflection & Lessons Learned

- Technical: trade-offs between imperceptibility, robustness, decentralisation.
- Interoperability challenges in blockchain and biometrics.
- Personal: learnt importance of privacy-by-design and ethical evaluation.
- Lesson: iterative prototyping and usability testing are critical..

Future Work

- Improve watermarking resilience to geometric attacks.
- Scale perceptual search (FAISS/HSNW ANN).
- Cross-chain provenance and interoperability.
- Multimodal biometrics (voice, behaviour).
- Support for video/audio provenance.
- Policy and regulator engagement.

Conclusion

- MyMark integrates watermarking, perceptual hashing, blockchain, and biometrics.
- Achieves provenance + identity binding without storing PII.
- Advances state-of-the-art in privacy-preserving media security.
- Key message: Provenance does not require surveillance.
- Thank you.